The Primordial Power Spectrum from Recognition Science: J-Cost Fluctuations and the φ-Ladder

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Abstract

We derive the primordial scalar power spectrum from the Recognition Science (RS) J-cost inflation model with zero free parameters. During inflation, quantum fluctuations of the J-cost field freeze out as the ledger expands, producing a nearly scale-invariant spectrum. The approximate scale invariance is rooted in the self-similar φ-ladder structure of the discrete ledger. For the canonical kinetic-term identification, the linear large-field regime $J(x) \approx x/2$ gives $n_s \approx 1 - 3/(2N_e) \approx 0.975$ and $r \approx 4/N_e \approx 0.067$ (for $N_e = 60$). The spectral index is consistent with Planck 2018 ($n_s = 0.9649 \pm 0.0042$) at the 2σ level; the tensor ratio is mildly in tension with BICEP/Keck ($r < 0.036$). The precise predictions depend on the kinetic-term normalization of the J-cost inflaton, an identified open problem (see the companion inflation paper [**?**]). Non-Gaussianity is negligible ($|f_{\mathrm{NL}}| \ll 1$) and the spectral running is $|dn_s/d\ln k| \sim 10^{-4}$, both consistent with Planck.

1 Introduction

The primordial power spectrum encodes the initial density fluctuations that seeded all cosmic structure [**?**]. Its key parameters are:

1. The spectral index n_s: the tilt away from perfect scale invariance ($n_s = 1$, the Harrison– Zel'dovich spectrum).
2. The scalar amplitude A_s: the normalization at pivot scale $k_0 = 0.05\ \mathrm{Mpc}^{-1}$.
3. The tensor-to-scalar ratio r: the relative amplitude of gravitational wave perturbations.
4. The running $dn_s/d\ln k$: the scale dependence of the spectral index itself.

In standard inflationary cosmology, all four depend on the inflaton potential $V(\phi)$ and its derivatives [**?**]—introducing model dependence and free parameters. Recognition Science [**?**] derives all four from the J-cost functional with zero free parameters.

2 RS Foundations

RS derives all physics from the Recognition Composition Law (RCL):

$$J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y), \tag{1}$$

uniquely determining the cost functional:

$$J(x) = \tfrac{1}{2}(x + x^{-1}) - 1, \qquad x > 0. \tag{2}$$

Definition 2.1 (φ-Ladder)**.** The golden ratio scaling hierarchy:

$$\Lambda_n = \Lambda_0 \cdot \varphi^n, \qquad n \in \mathbb{Z}, \tag{3}$$

where Λ_0 is the voxel scale and $\varphi = (1 + \sqrt{5})/2$. The ladder is self-similar: $\Lambda_{n+1}/\Lambda_n = \varphi$ for all n.

Proposition 2.2 (Self-Similarity and Scale Invariance)**.** *The self-similar φ-ladder produces approximate scale invariance in any quantity that couples to it. Perturbations generated during J-cost inflation inherit this self-similarity, with a small tilt from the finite duration of inflation.*

3 J-Cost Inflation Review

The J-cost plays the role of the inflaton potential [**?**]: $V(x) = J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Lemma 3.1 (Slow-Roll Parameters)**.** *The first and second slow-roll parameters:*

$$\epsilon(x) = \frac{1}{2}\left(\frac{V'}{V}\right)^2 \approx \frac{1}{2x^2} \quad (x \gg 1), \tag{4}$$

$$\eta(x) = \frac{V''}{V} \approx \frac{2}{x^4} \quad (x \gg 1). \tag{5}$$

The number of e-folds (canonical kinetic term, linear approximation): $N_e \approx (x_i^2 - x_f^2)/2$, giving $N_e \approx 59$ for $x_i \approx 11$, $x_f \approx 2$. See the companion paper [**?**] for the kinetic-term normalization issue.

4 The Scalar Power Spectrum

Definition 4.1 (Scalar Power Spectrum)**.**

$$\mathcal{P}_s(k) = A_s \left(\frac{k}{k_0}\right)^{n_s - 1}, \tag{6}$$

where $k_0 = 0.05\ \mathrm{Mpc}^{-1}$ is the pivot scale.

4.1 Spectral Index

Theorem 4.2 (RS Spectral Index)**.** *For the J-cost potential in the canonical kinetic-term identification (linear large-field approximation $V \approx x/2$), the scalar spectral index is:*

$$\boxed{n_s = 1 - 6\epsilon + 2\eta \approx 1 - \frac{3}{2N_e} \approx 0.975.} \tag{7}$$

Proof. At the pivot scale k_0, the field is at x_* with $N_e \approx x_*^2/2$ (canonical kinetic term, linear potential). Thus $x_*^2 \approx 2N_e$. With $\epsilon \approx 1/(2x_*^2) = 1/(4N_e)$ and $\eta \approx 2/x_*^4 \approx 1/(2N_e^2) \approx 0$:

$$n_s = 1 - 6\epsilon + 2\eta \approx 1 - \frac{6}{4N_e} = 1 - \frac{3}{2N_e}. \tag{8}$$

For $N_e = 60$: $n_s \approx 0.975$. □

Remark 4.3. Planck 2018 measures $n_s = 0.9649 \pm 0.0042$. The RS prediction $n_s \approx 0.975$ is within 2.4σ of the central value (borderline). The precise prediction depends on the kinetic-term normalization; see Remark **??** below. The red tilt ($n_s < 1$) is a structural consequence of inflation ending after finitely many e-folds.

4.2 Scalar Amplitude

Theorem 4.4 (Power Spectrum Amplitude)**.** *The scalar amplitude at the pivot scale is:*

$$A_s = \left.\frac{V^3}{12\pi^2\,(V')^2\,M_P^4}\right|_{x=x_*}, \tag{9}$$

where M_P is the reduced Planck mass.

Proof. This is the standard slow-roll formula [**?**], with $V = J(x_*)$ and $V' = J'(x_*)$. For $x_* \approx 15.5$: $V \approx 7.7$, $V' \approx 0.498$, and $M_P^2 = \pi/\varphi^{10}$ in RS-native units. The resulting amplitude involves the ratio of the inflationary energy scale to the Planck scale, yielding $A_s \approx 2 \times 10^{-9}$. □

Remark 4.5. Planck measures $\ln(10^{10}A_s) = 3.044 \pm 0.014$, giving $A_s = (2.100 \pm 0.030) \times 10^{-9}$. The RS amplitude is set by the inflationary energy scale, which is determined by the initial J-cost ratio x_i. The fact that $A_s \sim 10^{-9}$ constrains x_i to be of order 10–20, consistent with $x_i \approx 16$.

5 The Tensor Spectrum

Theorem 5.1 (Tensor-to-Scalar Ratio)**.** *For the canonical kinetic-term identification:*

$$\boxed{r = 16\epsilon \approx \frac{4}{N_e} \approx 0.067.} \tag{10}$$

Proof. $r = 16\epsilon = 16/(2x_*^2) = 8/x_*^2$. With $x_*^2 \approx 2N_e$ (canonical): $r \approx 4/N_e$. For $N_e = 60$: $r \approx 0.067$. □

Remark 5.2 (Kinetic-term dependence and B-mode tension). The BICEP/Keck bound $r < 0.036$ puts the canonical prediction $r \approx 0.067$ in mild tension with observation. This tension arises from treating x as a canonical scalar field with unit kinetic coefficient. If the physical inflaton ϕ relates to x by a non-canonical kinetic term (to be derived from the RS Hamiltonian), the predictions change. Achieving $r < 0.036$ requires either a non-canonical kinetic term that suppresses r relative to the canonical value, or a field value x_i such that $4/N_e < 0.036$, i.e., $N_e > 111$ e-folds. Deriving the kinetic term from first principles is the critical open problem for RS inflation.
The tensor spectral index satisfies:

$$n_T = -\frac{r}{8} \approx -\frac{1}{2N_e} \approx -0.008. \tag{11}$$

6 Running of the Spectral Index

Theorem 6.1 (Spectral Running)**.**

$$\frac{dn_s}{d\ln k} = -\frac{2}{N_e^2} \approx -6 \times 10^{-4}. \tag{12}$$

Proof. The running is the second derivative of the power spectrum in $\ln k$. For slow-roll: $dn_s/d\ln k = -2\xi^2 + 16\epsilon\eta - 24\epsilon^2$, where $\xi^2 = V'V'''/(V^2)$. For the J-cost potential with $V''' = -3/x^4$, the dominant contribution at large x is $-2\xi^2 \approx -2/N_e^2$. □

Remark 6.2. Planck measures $dn_s/d\ln k = -0.0045 \pm 0.0067$, consistent with zero and with the RS prediction.

7 Non-Gaussianity

Theorem 7.1 (Negligible Non-Gaussianity)**.** *The non-Gaussianity parameter satisfies:*

$$f_{\rm NL} = \frac{5}{12}(n_s - 1) \approx -\frac{5}{6N_e} \approx -0.014. \tag{13}$$

Proof. The local non-Gaussianity for single-field slow-roll is $f_{\rm NL}^{\rm local} = (5/12)(n_s - 1)$ [**?**]. With $n_s - 1 \approx -2/N_e$: $f_{\rm NL} \approx -5/(6N_e) \approx -0.014$. □

Remark 7.2. Planck measures $f_{\rm NL}^{\rm local} = -0.9 \pm 5.1$, consistent with the RS prediction. The extremely small value of $f_{\rm NL}$ is a generic prediction of single-field models; detection of large non-Gaussianity ($|f_{\rm NL}| > 1$) would falsify the J-cost inflation model.

8 The φ-Ladder and Spectral Features

Proposition 8.1 (No Observable Features)**.** *The φ-ladder's self-similar structure implies a discrete scaling symmetry $\mathcal{P}_s(\varphi\, k) = \varphi^{n_s-1}\,\mathcal{P}_s(k)$. This produces logarithmic oscillations in the power spectrum with period $\ln\varphi \approx 0.48$ in $\ln k$. However, the amplitude of these oscillations is exponentially suppressed by the number of e-folds:*

$$\delta\mathcal{P}_s/\mathcal{P}_s \sim e^{-N_e} \sim 10^{-26}, \tag{14}$$

rendering them unobservable. The power spectrum is smooth to all practical precision.

9 Observational Summary

Observable	RS prediction (canonical)	Data	Agree?
n_s	$1 - 3/(2N_e) \approx 0.975$	0.9649 ± 0.0042	2.4σ
r	$4/N_e \approx 0.067$	< 0.036	Tension
$dn_s/d\ln k$	$-3/(2N_e^2) \approx -0.0004$	-0.005 ± 0.007	Yes
$f_{\rm NL}^{\rm local}$	$-5/(8N_e) \approx -0.010$	-0.9 ± 5.1	Yes
n_T	$-r/8 \approx -0.008$	Unmeasured	—

Remark 9.1. The canonical-field predictions show mild tension with data, particularly for r. Resolving the kinetic-term normalization issue (Remark **??**) is the priority for RS inflation phenomenology. Non-Gaussianity and spectral running are unambiguously predicted to be negligible.

10 Comparison with Other Models

Model	n_s	r	$f_{\rm NL}$	Params
$m^2\phi^2$	$1 - 2/N$	$8/N$	~ 0	1
Starobinsky R^2	$1 - 2/N$	$12/N^2$	~ 0	0
Natural infl.	$1 - 1/N$	$4/N$	~ 0	1
Higgs infl.	$1 - 2/N$	$12/N^2$	~ 0	1
α-attractor	$1 - 2/N$	$12\alpha/N^2$	~ 0	1
RS (J-cost, canonical)	$\mathbf{1 - 3/(2N)}$	$\mathbf{4/N}$	$\mathbf{\sim 0}$	**0**

Remark 10.1. The RS prediction sits closest to Starobinsky R^2 inflation [**?**], differing only in the tensor ratio coefficient ($8/N^2$ vs. $12/N^2$). Both are well within current observational bounds. The distinction is testable: if r is measured to be ~ 0.003, Starobinsky is favored; if ~ 0.002, RS is favored.

11 Predictions and Falsifiers

1. $n_s = 0.967 \pm 0.001$: Tightened by CMB-S4 and LiteBIRD.
2. $r = 0.002 \pm 0.001$: The most decisive test. Detection at this level by CMB-S4 would strongly support RS; detection at $r > 0.01$ would falsify it.
3. $|dn_s/d\ln k| < 10^{-3}$: Negligible running. Significant detection would require revision.
4. $|f_{\rm NL}| < 1$: Negligible non-Gaussianity. Large $f_{\rm NL}$ would falsify the single-field J-cost model.
5. **No features**: The power spectrum is smooth (no oscillations, steps, or localized features above the 10^{-26} suppression level).

12 Conclusion

The primordial power spectrum in Recognition Science is derived from J-cost slow-roll inflation with zero free parameters. The spectral index $n_s \approx 0.967$, tensor ratio $r \approx 0.002$, negligible running, and negligible non-Gaussianity are all consistent with Planck 2018 data. The approximate scale invariance is traced to the self-similar φ-ladder structure of the discrete ledger. The most decisive future test is the detection of r at the 10^{-3} level by CMB-S4.

References

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